

Lectures 10: Generalized Estimating Equations (GEE) models/methods

April 10, 2025

Recall from last lecture: Fixed-Effect Models

- A good R package for the fixed-effect model is “plm” library
- The “plm” package can also do the random-effects model (although “lme4” might be better and more flexible).

A comparison between lme4 and plm

- **lme4**

- Linear Mixed Effects model (multilevel/hierarchical models with both fixed and random effects)
- `lmer(Y ~ X1 + X2 + (1|id), data)`

- **plm**

- Economic panel data, e.g., data with observations over time for the same units – e.g., individuals, countries
- `plm(Y ~ X1 + X2, index=c("id", "time"), data)`

See R code comparison!

This lecture

- Generalized Linear Model (**GLM**)
- Generalized Estimating Equations (**GEE**)
- R: “**geeglm**” in the “**geepack**” library

Recall Linear and logistic regression:

- **Data**: $(Y_i, X_{1i}, \dots, X_{pi})$ for $i = 1, \dots, m$
- **Models**: depending on the values of Y 's:
 - **Linear regression** when Y_i are continuous:
 - Assume $Y_i \sim N(\mu_i, \sigma^2)$ [or $Y_i = \mu_i + \epsilon_i, \epsilon_i \sim N(0, \sigma^2)$]
 - $\mu_i = \beta_0 + \beta_1 X_{1i} + \dots + \beta_p X_{pi}$
 - **Logistic regression** when Y_i are binary or binomial:
 - Assume $Y_i \sim \text{Bernoulli}(n_i, \pi_i)$ and
 - $\log \frac{\pi_i}{1-\pi_i} = \beta_0 + \beta_1 X_{1i} + \dots + \beta_p X_{pi}$

Generalized Linear Model (GLM)

- **Data:** $(Y_i, X_{1i}, \dots, X_{pi})$ for $i = 1, \dots, m$
- **Generalized Linear Model:** Two steps
 - Specify a distribution for response Y_i (say, with mean μ_i)
 - Specify a link function between μ_i and predictor variables:
$$g(\mu_i) = \beta_0 + \beta_1 X_{1i} + \dots + \beta_p X_{pi}$$
- **Three Special Cases:** depending on the values of Y 's:
 - When Y_i are continuous: **Linear regression:** assume $Y_i \sim N(\mu_i, \sigma^2)$ [or $Y_i = \mu_i + \epsilon_i, \epsilon_i \sim N(0, \sigma^2)$], and $g(\mu_i) = \mu_i$
 - When Y_i are binary or binomial: **Logistic regression.** assume $Y_i \sim \text{Bernoulli}(n_i, \pi_i)$ and $g(\pi_i) = \log \frac{\pi_i}{1-\pi_i}$
 - When Y_i are integer-valued count data: **Poisson regression.** Assume $Y_i \sim \text{Poisson}(\mu_i)$ and $g(\mu_i) = \log \mu_i$

The standard GLM

- **Data:** $(Y_i, X_{1i}, \dots, X_{pi})$ for $i = 1, \dots, m$
- The standard GLM:
 - The Y_i 's are **independent** with exponential family
 - A simpler one-parameter exponential family is
$$f_{\theta}(y) = f_0(y) \exp(\theta y - b(\theta))$$
where **$E_{\theta}(Y) = b'(\theta)$ and $Var_{\theta}(Y) = b''(\theta)$** under reasonable regularity conditions
 - Using the **maximum likelihood function (MLE)**: maximizes
$$\prod_{i=1}^m f_{\theta}(Y_i) = \prod_{i=1}^m f_0(Y_i) * \exp\left(\sum_{i=1}^m (\theta_i Y_i - b(\theta_i))\right)$$
where $g(\mu_i) = g(b'(\theta_i)) = \beta_0 + \beta_1 X_{1i} + \dots + \beta_p X_{pi}$

From Independent to Dependent

- The standard GLM assumes that the data are independent!
- What happens if the data can be dependent?
E.g., longitudinal data:
 - **Data:** $(ID, Y_{ij}, X_{1ij}, \dots, X_{pij})$ for $i = 1, \dots, m$ and $j = 1, \dots, n_i$
 - **Generalized Estimating Equations (GEE)**

History of The GEE Models

A Brief introduction to **Generalized Estimating Equations (GEE)** models/methods

- It was first proposed by Liang and Zeger (Biometrika, 1986)
 - Title: Longitudinal data analysis using generalized linear models.
 - Citation #: **22,255** as for April 2025
- It is semi-parametric model

Longitudinal Data: Vector notation

- **Data:** $(ID, Y_{ij}, X_{1ij}, \dots, X_{pij})$ for $i = 1, \dots, m$ and $j = 1, \dots, n_i$

Vector/Matrix notation **when $n_i \geq 2$** :

- $Y_i = \begin{pmatrix} Y_{i1} \\ \cdots \\ Y_{i,n_i} \end{pmatrix}, X_{1i} = \begin{pmatrix} X_{1i1} \\ \cdots \\ X_{1i,n_i} \end{pmatrix}, \dots, X_{pi} = \begin{pmatrix} X_{pi1} \\ \cdots \\ X_{pi,n_i} \end{pmatrix}$
- **Following the GLM model**, we assume **mean structure** :
 - Linear regression : $E(Y_i) = \mu_i = \beta_0 + \beta_1 X_{1i} + \dots + \beta_p X_{pi}$
 - Logistic regression: $\log \frac{\mu_i}{1-\mu_i} = \beta_0 + \beta_1 X_{1i} + \dots + \beta_p X_{pi}$
 - Or more generally, $g(\mu_i) = \beta_0 + \beta_1 X_{1i} + \dots + \beta_p X_{pi}$

Longitudinal Data

- **Data:** $(ID, Y_{ij}, X_{1ij}, \dots, X_{pij})$ for $i = 1, \dots, m$ and $j = 1, \dots, n_i$

Vector/Matrix notation **when $n_i \geq 2$** :

- $Y_i = \begin{pmatrix} Y_{i1} \\ \dots \\ Y_{i,n_i} \end{pmatrix}, X_{1i} = \begin{pmatrix} X_{1i1} \\ \dots \\ X_{1i,n_i} \end{pmatrix}, \dots, X_{pi} = \begin{pmatrix} X_{pi1} \\ \dots \\ X_{pi,n_i} \end{pmatrix}$

- **Variance / Correlation structure:**

- $\text{Var}(Y_i) = \Omega_i \quad [n_i \times n_i]$

- For intercept-random-effects linear mixed-effects model:

$$Y_{ij} = \beta_0 + \beta_1 X_{1i} + \dots + \beta_p X_{pi} + b_i + \epsilon_{ij}, \text{ we have}$$

$$\Omega_i = \begin{pmatrix} \sigma_b^2 + \sigma_\epsilon^2 & \dots & \sigma_b^2 \\ \vdots & \ddots & \vdots \\ \sigma_b^2 & \dots & \sigma_b^2 + \sigma_\epsilon^2 \end{pmatrix} = \sigma_\epsilon^2 \begin{pmatrix} 1/\theta^2 + 1 & \dots & 1/\theta^2 \\ \vdots & \ddots & \vdots \\ 1/\theta^2 & \dots & 1/\theta^2 + 1 \end{pmatrix}$$

Parameter Estimation

Vector/Matrix notation **when** $n_i \geq 2$:

- $Y_i = \begin{pmatrix} Y_{i1} \\ \cdots \\ Y_{i,n_i} \end{pmatrix}, X_{1i} = \begin{pmatrix} X_{1i1} \\ \cdots \\ X_{1i,n_i} \end{pmatrix}, \cdots, X_{pi} = \begin{pmatrix} X_{pi1} \\ \cdots \\ X_{pi,n_i} \end{pmatrix}$
- **Mean structure:** $E(Y_i) = \mu_i = \beta_0 + \beta_1 X_{1i} + \cdots + \beta_p X_{pi}$
- **Variance/Correlation structure:** $\text{Var}(Y_i) = \boldsymbol{\Omega}_i \ [n_i \times n_i]$
- How to estimate the parameter $(\beta_0, \cdots, \beta_p)$?

MLE when the $\boldsymbol{\Omega}_i$'s are known (up to a constant)

- $Y_i = \begin{pmatrix} Y_{i1} \\ \cdots \\ Y_{i,n_i} \end{pmatrix}, X_{1i} = \begin{pmatrix} X_{1i1} \\ \cdots \\ X_{1i,n_i} \end{pmatrix}, \cdots, X_{pi} = \begin{pmatrix} X_{pi1} \\ \cdots \\ X_{pi,n_i} \end{pmatrix}$
- **Mean structure:** $E(Y_i) = \mu_i = \beta_0 + \beta_1 X_{1i} + \cdots + \beta_p X_{pi}$
- **Variance/Correlation structure:** $\text{Var}(Y_i) = \boldsymbol{\Omega}_i \quad [n_i \times n_i]$
- **For linear regression:** minimize the

$$WSS = \sum_{i=1}^m (Y_i - \mu_i)^T \boldsymbol{\Omega}_i^{-1} (Y_i - \mu_i)$$

- Taking derivatives with respect to a given β_h yields

$$\sum_{i=1}^m \frac{\partial \mu_i}{\partial \beta_h} \boldsymbol{\Omega}_i^{-1} (Y_i - \mu_i) = 0$$

Questions

- What if **Variance/Correlation structure:**
 $Var(Y_i) = \Omega_i$ is unknown?
- **Solution:**
Generalized Estimating Equations (GEE):
Replace Ω_i by a working matrix $\sigma^2 W_i$

GEE Model/Methods

Key assumptions of GEE:

- We know the mean structure:
 - Linear regression : $E(Y_i) = \mu_i = \beta_0 + \beta_1 X_{1i} + \dots + \beta_p X_{pi}$
 - Logistic regression: $\log \frac{\mu_i}{1-\mu_i} = \beta_0 + \beta_1 X_{1i} + \dots + \beta_p X_{pi}$
 - Or more generally, $g(\mu_i) = \beta_0 + \beta_1 X_{1i} + \dots + \beta_p X_{pi}$
- $V(\mu)$ = mean variance relationship. For example, GEE logistic regression:
$$\text{Var}(Y) = \phi \mu(1 - \mu).$$
- We have a **working correlation matrix** W_i for $Y_i = \begin{pmatrix} Y_{i1} \\ \vdots \\ Y_{i,n_i} \end{pmatrix}$ which can be different from true ***corr***(Y_i).
 - **It is okay to mis-specify the covariance-variance matrix!!!**
 - **W_i = working correlation structure** such as “exchangeable”, “AR”, etc.
- **Our primary interests are to estimate** regression coefficients $(\beta_0, \dots, \beta_p)$!

Key ideas: score-like equations

- For MLE, score equations are formulated by setting the partial derivatives of the log-likelihood to zero for each unknown parameter.
- In GLM, for each h ,

$$\sum_{i=1}^m \frac{\partial \mu_i}{\partial \beta_h} [Var(Y_i)]^{-1} [Y_i - \mu_i] = 0$$

- The form of GEE is similar: given the variance-covariance matrix W_i (or working covariance matrix)

$$\sum_{i=1}^m \frac{\partial \mu_i}{\partial \beta_h} [W_i]^{-1} [Y_i - \mu_i] = 0$$

Effects of mis-specifying correlation structure

- Not usually so striking
- Standard error estimates more often affected than parameter estimates
- Possible effects on both standard error and parameter estimates

Interpretation of GEE Model

When using the GEE approach to carry out logistic regression for correlated dichotomous responses, the interpretation of GEE model is similar to standard logistic regression

- Similarity: use same formulas for
 - Point estimates
 - OR estimates
 - Confidence intervals
 - (Wald) test statistics
- Difference
 - Underlying assumptions
 - Method of parameter estimation

Correlation Structures

- The most common correlation structures are:
 - **Independence:**
 - Correlation between time points is independent
 - **Exchangeable:**
 - Same correlation for all time pairs
 - **Autoregressive:**
 - Correlation declines exponentially with the time lag
 - This one assumes that there is a constant time interval between occasions
 - **Unstructured:**
 - There can be different correlations for each pairs of data
 - Used often for longitudinal data where there are not many time points

Correlation Structures

How do I decide which structure to use with longitudinal data?

- Start with exchangeable and if it looks like there is some type of pattern then try a different structure.
- Also, if there is some reason to believe that you should use another structure (e.g. previous work, statistical theory), do so.

More discussions of GEE

- There are more restrictive correlation structures but they are very rarely used
- GEE can handle missing and unbalanced data under the assumption that data are missing completely at random (MCAR)
- We specify robust standard errors because it assumes that the correlation will be the same across time
 - This will provide approximately unbiased estimates, but not necessarily efficient
 - GEE produces efficient estimates

Comparison: GEE vs. Other Models

- Random effect models: `lmer(Y ~ X1 +..+Xp + (Z|id) )`
 - Robust standard errors (produced unbiased estimates)
 - Did not discuss in class much, but important when thinking about population average models
 - Popular in economics and political science
- Fixed effect models: `lm(Y ~ factor(id) + X1..+Xp)`
 - Assessing the (causal) relationship between a time-vary predictor and outcome
 - Need at least 3 time points
 - Must have significant variation between timepoints
- GEE (marginal model): `geeglm(Y~ X1+...+Xp, id, corstr)`
 - Tells us what is happening among participants
 - Use with robust standard errors
 - Specify correlation matrix

Additional GEE Resources

- <https://online.stat.psu.edu/stat504/node/180/>
- <https://rlbarter.github.io/Practical-Statistics/2017/05/10/generalized-estimating-equations-gee/>
- <https://www.stata.com/features/generalized-estimating-equations/>

Next Class...

- A brief introduction to Polynomials
- Count outcomes
 - Poisson, Negative binomials, zero-inflated models
- Other models
 - Group based trajectory analyses

Administrative Stuff

- **Assignment 4**
 - Due 4/11 11:59pm
- Group presentation discussion

Group Project Presentation

- On average, 6 minutes per person, e.g.,
 - 12 minutes for 2-people team and
 - 18-minutes for 3-people team
- Background (1~2 slide)
 - Research question
 - Variables selected
- Results (about 3~6 slides)
 - Tables/Graphs
- Conclusion (1~2 slide)